Different Approach but discriminative in nature:

General idea of Baye's theorem

$$P(Y/x) = \frac{P(x/y) \cdot P(y)}{P(x)}$$

$P(x/y)$ = likelihood

$P(y)$ = Prior

Assuming we have a sequence of input words $X = x_1, x_2 \ldots x_n$ and want to compute a sequence of output tags $Y = y_1, y_2 \ldots y_n$.

HMM Approach

We maximize $P(Y/x)$, by relying on Baye's rule and the likelihood $P(x/y)$

$$\hat{y} = \underset{y}{\operatorname{argmax}} P(Y/x)$$

$$= \underset{y}{\operatorname{argmax}} P(x/y) \cdot P(y)$$

$$= \underset{y}{\operatorname{argmax}} \underbrace{\prod_i P(x_i/y_i)}_{\text{Emission Probs}} \cdot \underbrace{\prod_i P(y_i/y_{i-1})}_{\text{Transition Probs}}$$

In a CRF (conditional random fields), by contrast, we compute the positerior $P(y/x)$ directly.

Step 1

Goal

$$\hat{y} = \underset{y}{\operatorname{argmax}} \; P(y/x)$$

CRF won't compute the probability for each tag at each time stamp. Instead, at each time CRF computes log-linear functions over a set of relevant features, these local features are aggregated and normalized to produce a global probability for the whole sequence

Step2 :

Conditional probability

$$P(y/x) = \frac{\exp\left(\sum_{k=1}^{K} w_k f_k(x,y)\right)}{\sum_{y' \in y} \exp\left(\sum_{k=1}^{K} w_k F_k(x,y')\right)}$$

$$P(y/x) = \frac{1}{Z(x)} \exp\left(\sum_{k=1}^{K} w_k f_k(x,y)\right)$$

$$\boxed{F_k(x,y) = \sum_{i=1}^{n} f_k(y_{i-1}, y_i, x, i)}$$ Global features

$$P(y/x) = \frac{1}{Z(x)} \exp\left(\sum_{i=1}^{n}\sum_{k=1}^{K} w_k f_k(y_{i-1}, y_i, x, i)\right)$$

Step3

Feature functions

Each f_k is a binary (or) real-valued function

Capturing a real useful pattern.

Examples:

- $f_1(y_{i-1}, y_i, x, t) = 1$ if $y_{i-1} = DET$, $y_i = NOUN$
- $f_2(y_i, x, t) = 1$ if $y_i = VERB$ and x_i ends in '-ing'

Then we weigh these features using w_k, indicating how important they are.

Step4:

Training MLE

We want to maximize the log likelyhood over the training set

$$L(w) = \sum_{n=1}^{N} \log\left(P\left(y^{(n)}/x^{(n)}\right)\right)$$

Using CRF

$$\log(P(y/x)) = \left[\sum_{i=1}^{N}\sum_{k=1}^{K} w_k f_k(y_{i-1}, y_i, x, i) - \log(Z(x))\right]$$

$$L(w) = \sum_{n=1}^{N} \left[\sum_{i=1}^{n} \sum_{k=1}^{K} w_k f_k (y_{i-1}^{(n)}, y_i^{(n)}, x^{(n)}, i) - \log z(x^{(n)}) \right]$$

<u>Step 5:</u>
Gradient

$$\frac{\partial L}{\partial w} = \sum_{n=1}^{N} \left(\sum_{i=1}^{n} f_k (y_{i-1}^{(n)}, y_i^{(n)}, x^{(n)}, i) - \sum_{p(y/x^{(n)})} \left[f_k (y_{i-1}, y_i, x^{(n)}, i) \right] \right)$$

f_k is observed in the training data

Expected count of f_k under the model's Current distribution

<u>Step 6:</u>
Inference

$$\hat{y} = \underset{y}{\arg\max} \left(P(y/x) \right)$$

$$\hat{y} = \underset{y}{\arg\max} \sum_{i=1}^{n} \sum_{k=1}^{K} \lambda_k f_k (y_{i-1}, y_i, x, i)$$

Maximize the feature score sum

we can implement this using <u>Viterbi Algo</u>

Example :

Input

$x = ($ "fish", "swim" $)$

Possible tags

$y = ($ "VERB", "NOUN" $)$

Goal is to compute $p(y/x)$

1. Define features

Define 2 simple feature functions

Transition

$$f_1(y_{i-1}, y_i, x, i) = \begin{cases} 1 & \text{if } y_{i-1} = \text{Noun}, y_i = \text{Verb} \\ 0 & \text{Otherwise} \end{cases}$$

Emission

$$f_2(y_i, x, i) = \begin{cases} 1 & \text{if } y_i = \text{noun and } x_i = \text{fish} \\ 0 & \text{otherwise} \end{cases}$$

assign weights

$w_1 = 1.0$ (transition)

$w_2 = 2.0$ (emission)

Step 2: List all possible label sequences

$x = (x_1, x_2)$ and each $y_i \in$ (noun, verb)

(noun, noun)
(noun, verb)
(verb, noun)
(verb, verb)

Step 3: Scores

$$\text{Score}(y, x) = \exp\left(\sum_{i=1}^{2} \sum_{K=1}^{2} w_K f_K(x) \right)$$

noun	noun	$\exp(1.0 + 2.1)$	$= 7.389$
noun	verb	$\exp(1.1 + 2.1)$	$= 20.086$
verb	noun	$\exp(0)$	$= 1.0$
verb	verb	$\exp(0)$	$= 1.0$

Step 4: Partition function

$$Z(x) = \sum_{y'} \text{score}(y', x) = 7.389 + 20.086 + 1 + 1$$
$$= 29.475$$

Step5: Compute Probabilities

$$P(N,N) = 0.251$$
$$P(V,N) = 0.034$$
$$P(N,V) = 0.681$$
$$P(V,V) = 0.034$$

Step6: Decode best sequence

$$\hat{y} = \text{argmax } P(y/x) = (\text{Noun, Verb})$$